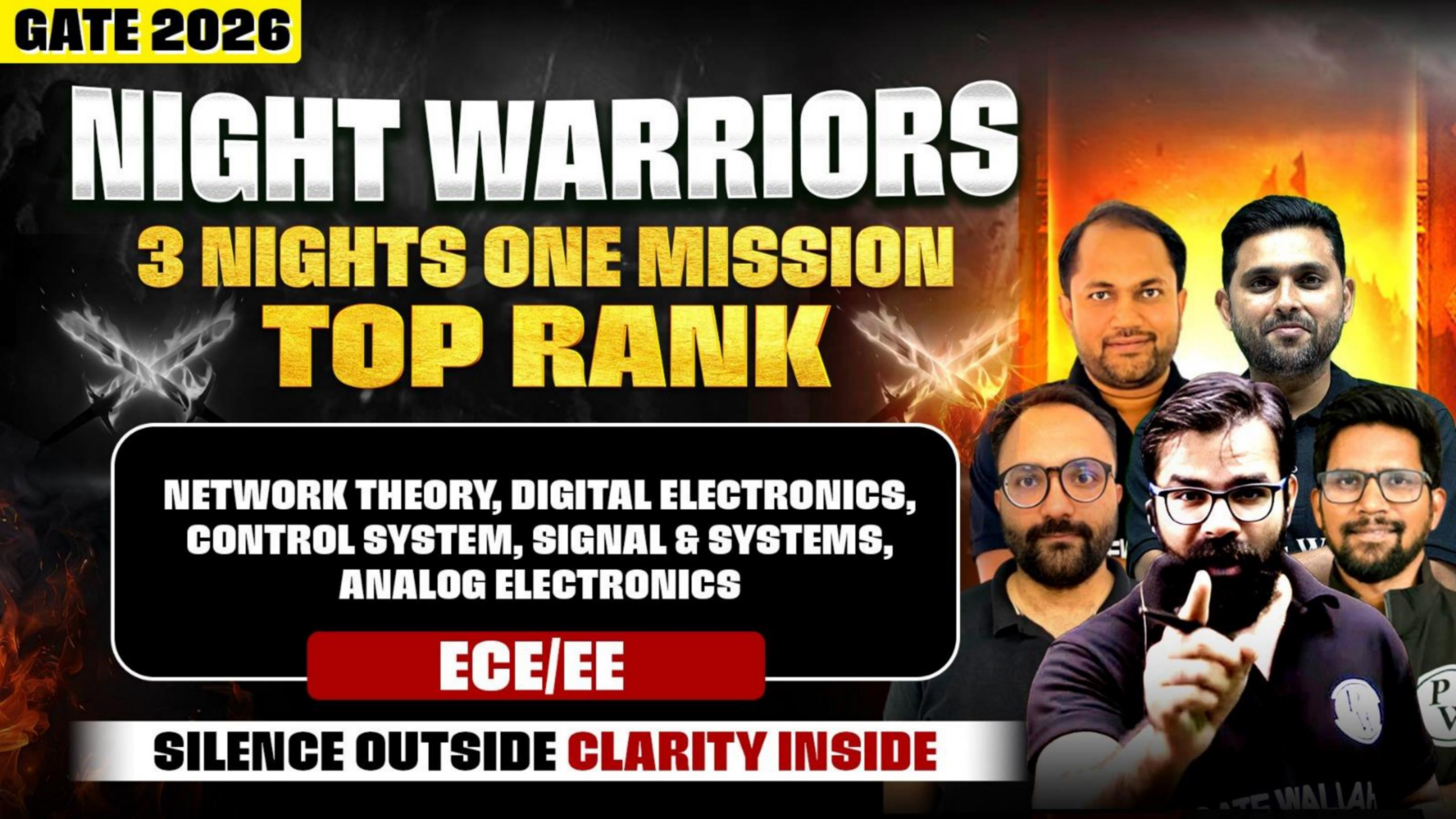

GATE 2026
NIGHT WARRIORS
3 NIGHTS ONE MISSION
TOP RANK
NETWORK THEORY, DIGITAL ELECTRONICS,
CONTROL SYSTEM, SIGNAL & SYSTEMS,
ANALOG ELECTRONICS
ECE/EE
SILENCE OUTSIDE CLARITY INSIDE

Question

#Q. Find the number of combination of A, B, C for which Y will be 1.

(A) 4

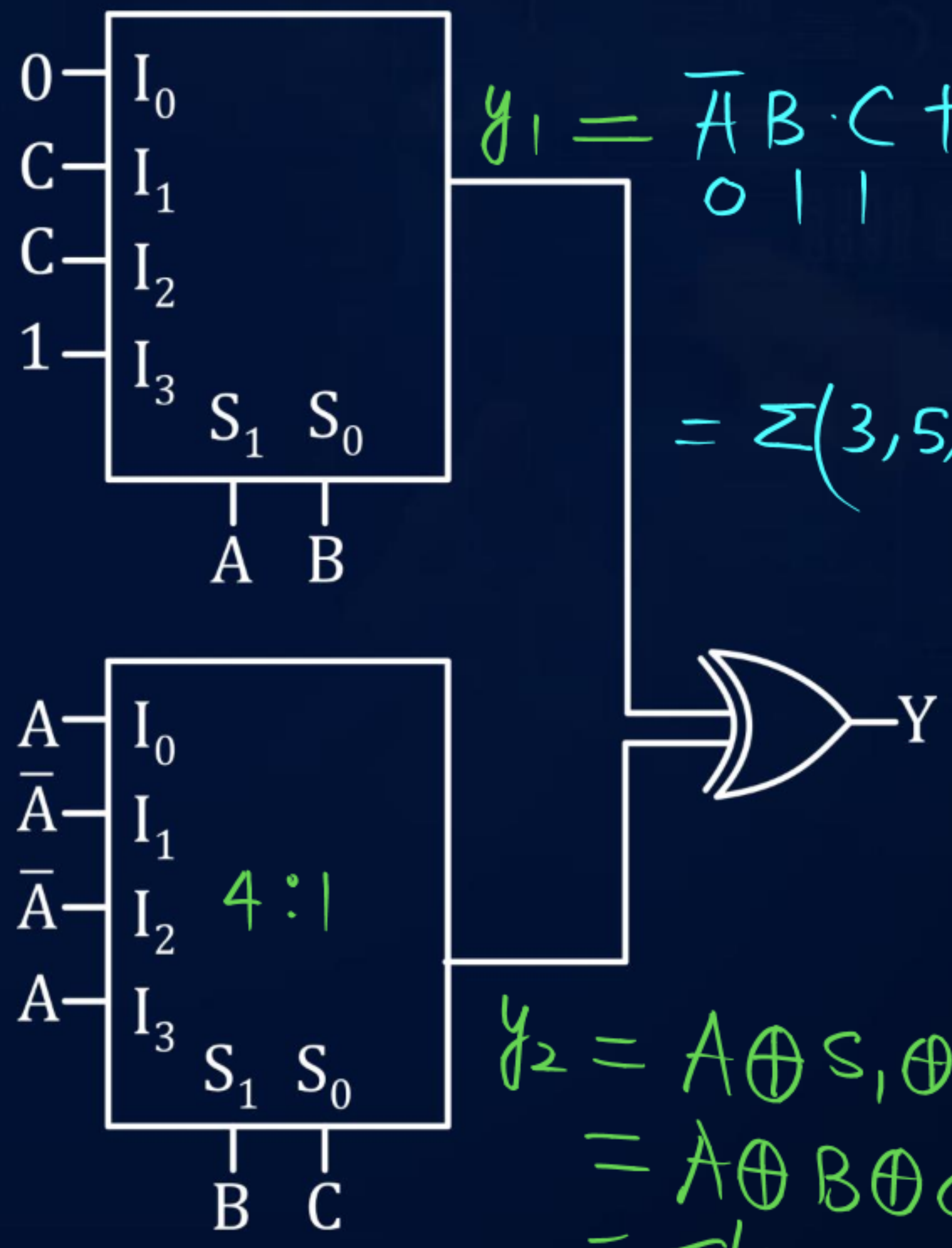
(B) 5

(C) 6

(D) 7

$y = y_1 \oplus y_2$

$= \Sigma(1, 2, 3, 4, 5, 6)$



$y_1 = \bar{A} B \cdot C + A \bar{B} \cdot C + AB \cdot 1$

(1 0 1; 0 1 1)

$= \Sigma(3, 5, 6, 7)$

$110 \to 6$

$111 \to 7$

$y_2 = A \oplus S_1 \oplus S_0$

$= A \oplus B \oplus C$

$= \Sigma(1, 2, 4, 7)$

Question MSQ.

#Q. Find the correct equation/s.

(A) $AB + BC + AC = AB \oplus BC \oplus AC$

(A, B)

(B) $\bar{A} \oplus \bar{B} \oplus AB = A + B$ $\rightarrow$ $A \oplus B \oplus AB = (A+B)$

(C) $AB \odot BC = B(A \odot C)$

$AB \oplus BC = B.[A \oplus C]$

(D) $AB + BC + \bar{A}D + \bar{B}D + CD = \overline{(A + B)}D + AB + BC$

$= \bar{A}\bar{B}D + AB + BC$

$P = AB$ $\quad AB + BC + D(\bar{A} + \bar{B}) + CD$

$\Rightarrow$

$(P + \bar{P})(P + D) + BC + CD = AB + D + BC + CD = AB + BC + D$

Question

#Q. For the 8-bit flash type ADC, the number of resistor required is $\left(\frac{2^{10}}{2^x}\right)$. The value of x is 2 ________

8 bit $\longrightarrow$ Resistances $\longrightarrow$ $2^8 = \frac{2^{10}}{2^x}$

$2^x = 2^2$

$x = 2$

Question MSQ

#Q. For the following K-map. Find the minimized Boolean function/s.

A $\bar{A}D + BC + \bar{B}\bar{C}\bar{D} + A\bar{C}\bar{D}$

B $\bar{A}D + BC + A\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}$

C $\bar{A}D + BC + \bar{B}\bar{C}\bar{D} + AB\bar{D}$

D $\bar{A}D + BC + \bar{B}\bar{C}\bar{D} + A\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	
$\bar{A}B$		1	1	1
AB	1		1	1
$A\bar{B}$	1			

Question MSQ

#Q. Find the correct statement/s. (A, B, C, D)

A. $A\bar{B} \oplus BC = A\bar{B} + BC$

$A\bar{B} \oplus BC = A\bar{B} + BC$

B. $A \oplus \bar{B} \oplus AB = \overline{AB}$

$A \oplus \bar{B} \oplus A \cdot B = \overline{A \oplus B \oplus AB}$

$= \overline{A+B} = \bar{A} \cdot \bar{B}$

C. Boolean function, $A \oplus B \oplus \bar{A}B$, can be implement by minimum 3 NAND gate (2-i/p gate are used).

$A \oplus B \oplus \bar{A} \cdot B = \overline{\bar{A} \oplus B \oplus \bar{A}B}$

$= \overline{\bar{A}+B}$

$= A \cdot \bar{B}$ (3 NAND)

D. $\bar{A}C \oplus A\bar{C} = (\bar{A} + \bar{C})(A + C)$

$\bar{A}C \oplus A\bar{C} = \bar{A}C + A\bar{C} = A \oplus C$

$= (\bar{A}+\bar{C})(A+C)$

Question

#Q. Find the summation of PI and EPI terms in the following K-map.

(A) 6

(B) 8

(C) 10

(D) 12

PI → 6

EPI → 2

WX \ YZ	$\bar{Y}\bar{Z}$	$\bar{Y}Z$	YZ	$Y\bar{Z}$
$\bar{W}\bar{X}$	1	1		
$\bar{W}X$		1	1	
WX			1	1
$W\bar{X}$				1

Question

#Q. If roots of an equation $x^2 - 10x + 26 = 0$ are 4 and 7 then base is $\underline{(11)}$.

$$x^2 - (\alpha + \beta)x + \alpha \cdot \beta = 0$$

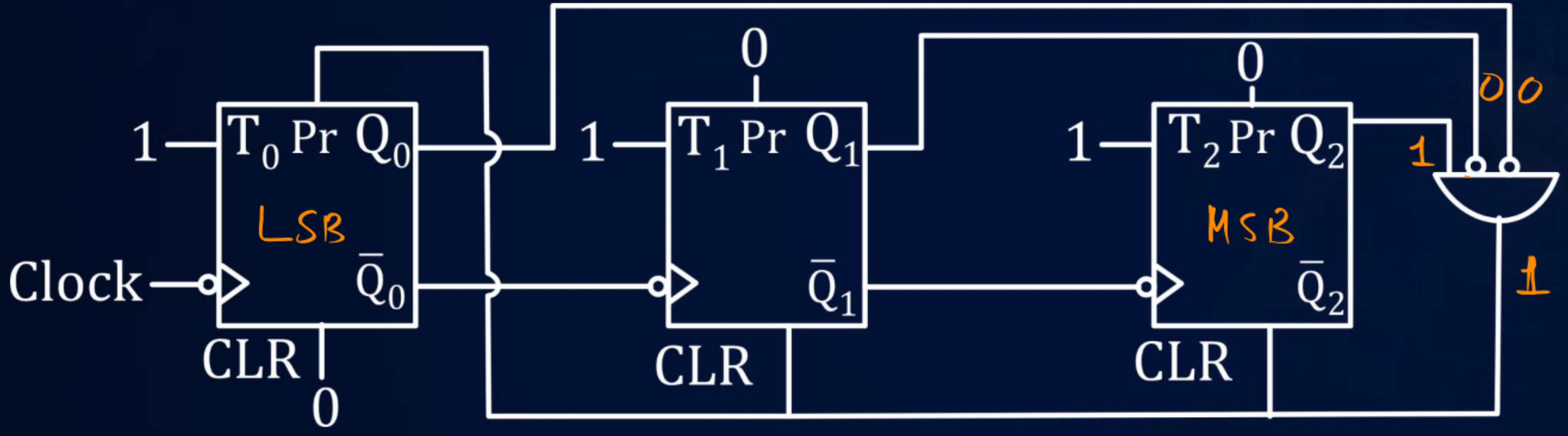
$$(\alpha)_r + (\beta)_r = (10)_r$$

$$\begin{array}{r} (4)_{11} \\ +\ (7)_{11} \\ \hline (10)_{11} \end{array}$$

Question

#Q. The MOD. no. of the given Counter is ___5___.

down counter



0 0 1

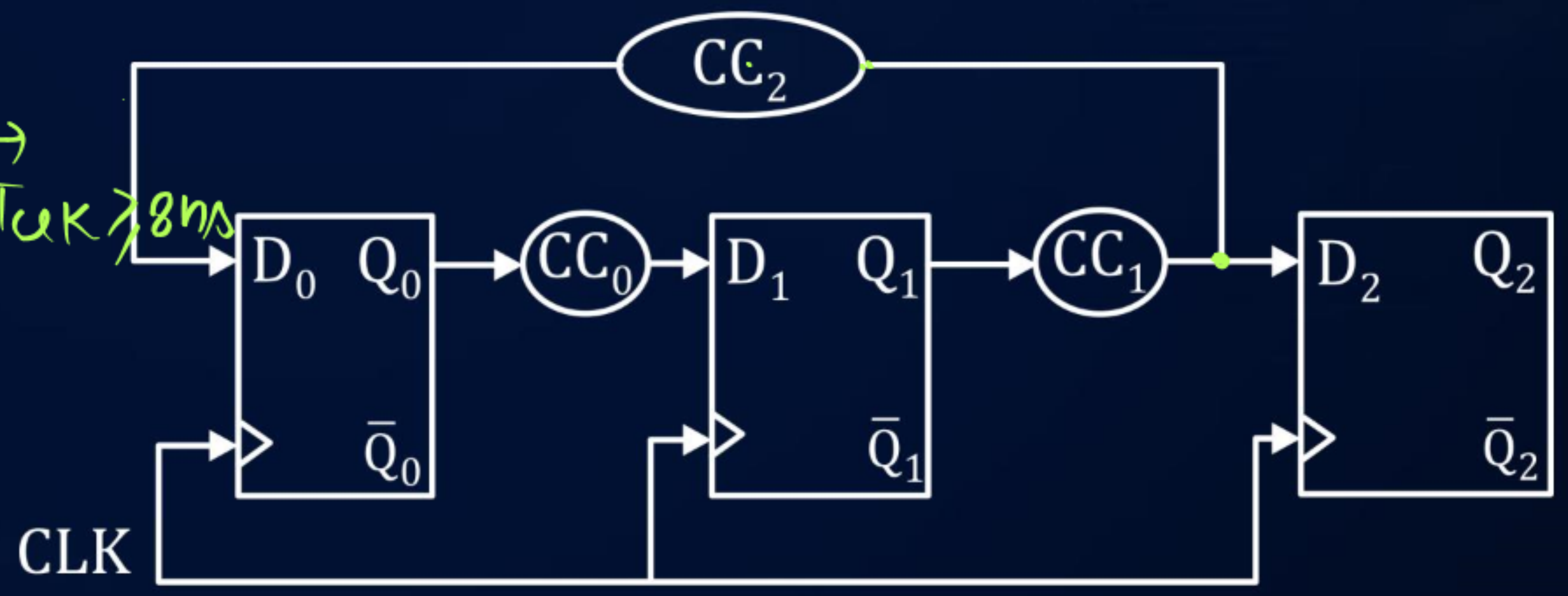
Q_2	Q_1	Q_0
0	0	0
1	1	1
1	1	0
1	0	1
1	0	0

MOD no. 5.

Question

#Q. Find the value of f_{max} for proper working. $t_{set\text{-}up} = t_{hold} = 1$ ns. CC stands for combinational circuit.

Circuit	Td
Flip-Flop	1ns
CC_0	2ns
CC_1	6 ns
CC_2	8 ns



(A) 20 MHz

(B) 33.33 MHz

(C) 62.5 MHz

(D) 67.67 MHz

Q_0 D_0 next trigger

1ns 14 ns 1ns

$T_{CLK} \geq 16$ns

1ns Q_1 D_1 → $T_{CLK} \geq 4$ ns

2ns 1ns

1ns Q_2 D_2 → $T_{CLK} \geq 8$ns

6ns 1ns

overall $T_{CLK} \geq 16$ns

$f_{CLK} \leq \frac{1}{16\text{ns}}$

$(f_{CLK})_{max} = \frac{1}{16\text{ns}} = \frac{10^3}{16}$ MHz

Question

#Q. P, Q are the integers corresponding to the 4-bit binary number 1110 considered in 1's complement and 2's complement representations, respectively. The 5-bit 1's complement representation of (Q - P) is?

(A) 11111

(B) 11011

(C) 11110

(D) 11100

$P = (1110)_2 \xrightarrow{1's} (0001)_2$

$P = (-1)_{10} \qquad (+1)_{10}$

$00001 \xrightarrow{1's} (11110)_2$

$(+1)_{10} \qquad (-1)_{10}$

$Q = (1110)_2 \xrightarrow{2's} (0010)_2$

$(-2)_{10} \qquad (+2)_{10}$

$Q - P = -2 - (-1)$

$= (-1)_{10}$

Question

#Q. Consider the signed binary number $\boxed{A = 01010110}$ and $\boxed{B = 11101100}$ where B is the 1's complement and MSB is the sign bit, then the value of $\boxed{-A\ -B}$ in 2's complement format is

$$\boxed{(-A)+(-B)}$$

(A) 10111100

(B) 10111101

(C) 01000010

(D) None of the above

$$A = (01010110)_2 \xrightarrow{2's} (10101010)_2 = -A$$

$$B = (11101100)_2 \xrightarrow{1's} (00010011)_2 = (-B)$$

$$(00010011)_2 \xrightarrow{2's} \underset{B \text{ in } 2's}{(11101101)_2} \xrightarrow{2's} -B = (00010011)_2$$

$$\begin{array}{r} 10101010 \\ +\ 00010011 \\ \hline (10111101)_2 \end{array}$$

Question

#Q. Which of the following notations have same representation of +0 and –0?

(A) 1's complement with radix of number being 2

(B) 7's complement with radix of number being 8

(C) 9's complement with radix of number being 10

(D) 10's complement with radix of number being 10

H.W

Question

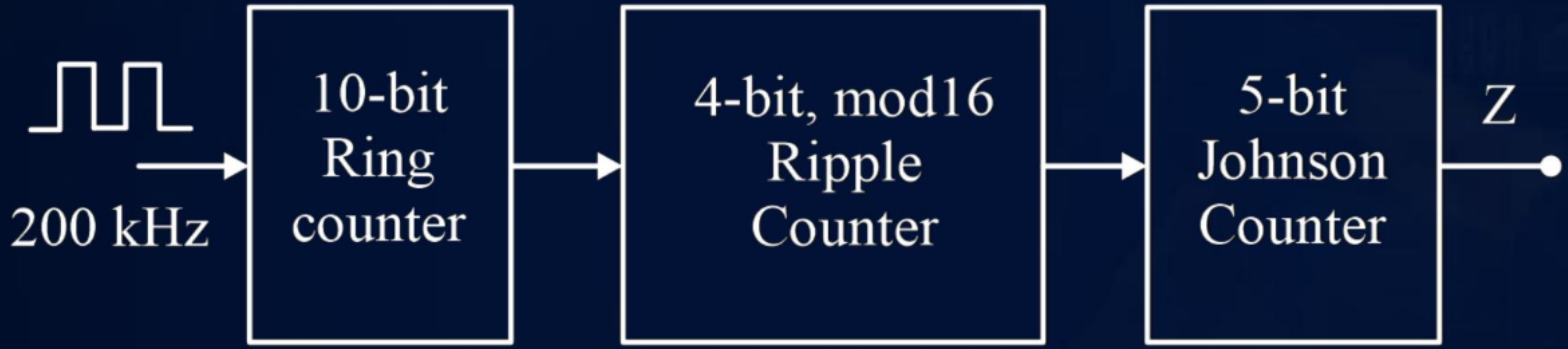
#Q. The frequency at the output of the following cascaded circuit is ________ Hz

A) 100

B) 125

C) 150

D) 200



H.W

Question

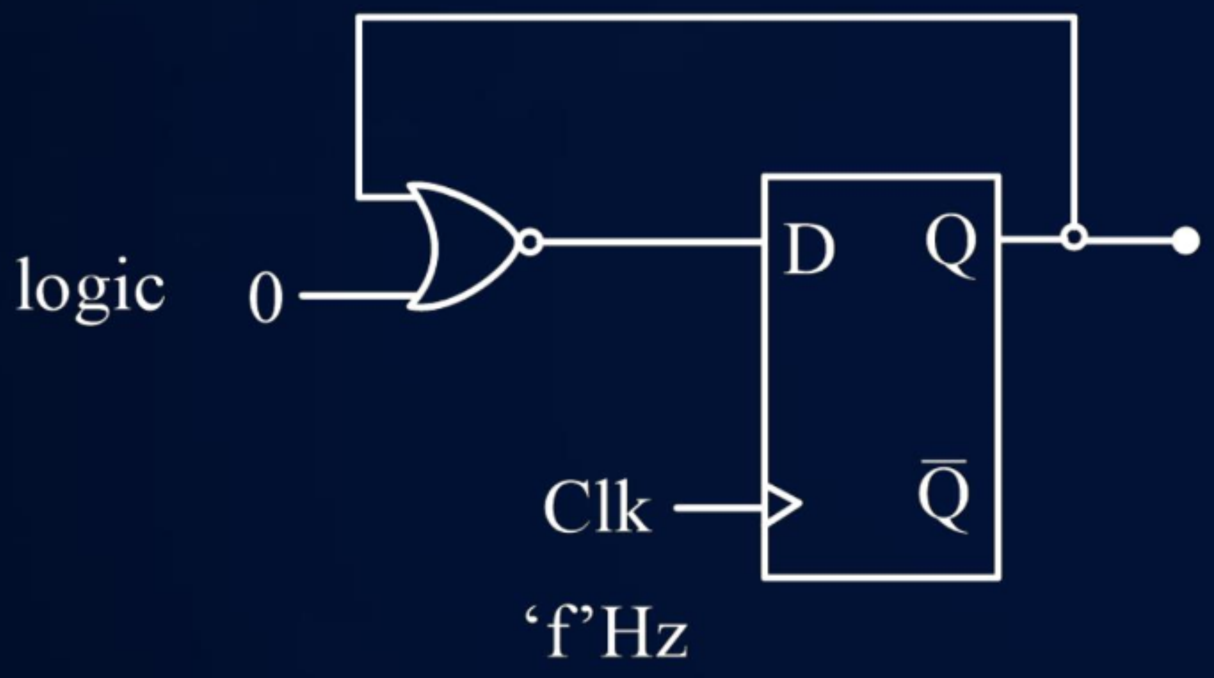
#Q. For the circuit shown below. If the clock frequency is 'f' Hz then frequency of the output Q is

(A) f

(B) 2f

(C) f/2

(D) f/3



H.W

Question

#Q. The mod number of the given counter

Assume initially counter is let to 1011.

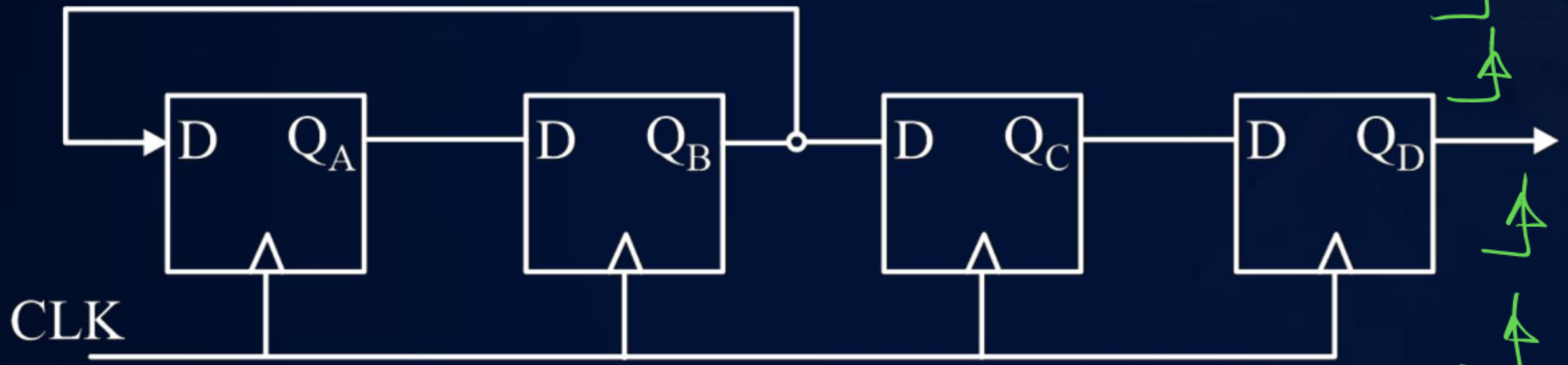
(A) 2

(B) 4

(C) 6

(D) 8

$Q_A(n+1) = D_A = Q_B(n)$



Q_A	Q_B	Q_C	Q_D
1	0	1	1
0	1	0	1
1	0	1	0
0	1	0	1
1	0	1	0

MOD no. = 2

Question

#Q. Consider the following shift right register.

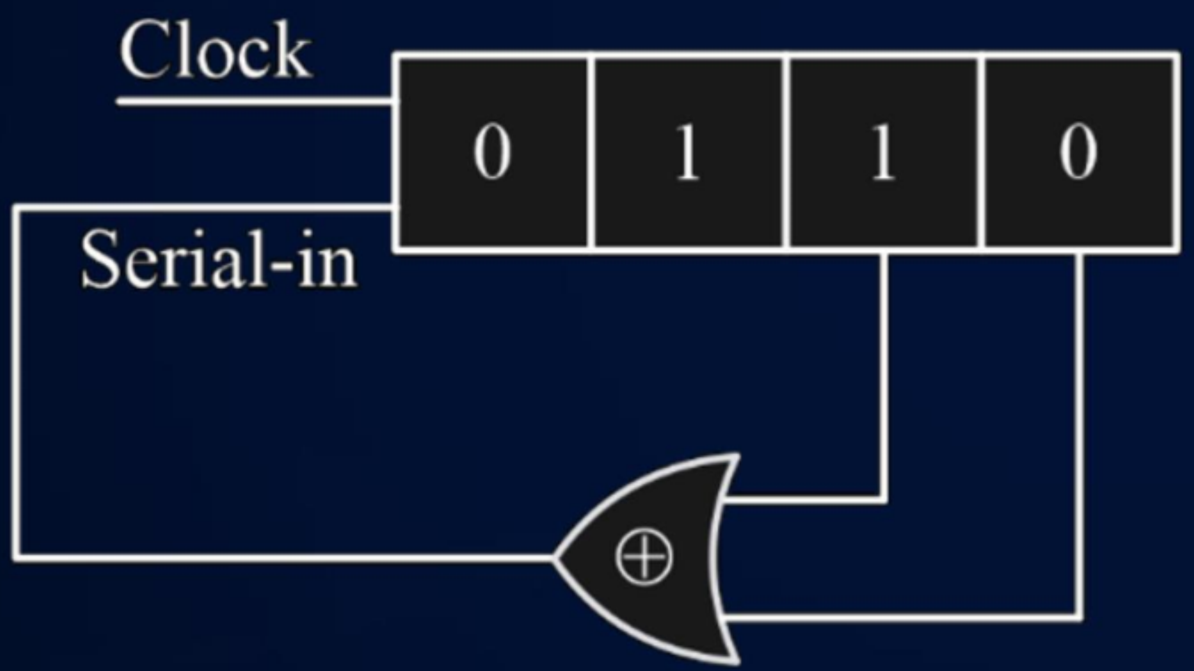
The initial contents of the 4-bit serial-in parallel out, shift right register shown above are 0110. What will be the contents of the register after 3 clock pulses ?

(A) 0000

(B) 0101

(C) 1010

(D) 1111



H.W

Question

#Q. The clock frequency applied to the digital circuit shown in figure below is 50 kHz. If initial state of the flipflop output 'Q' is '0'.

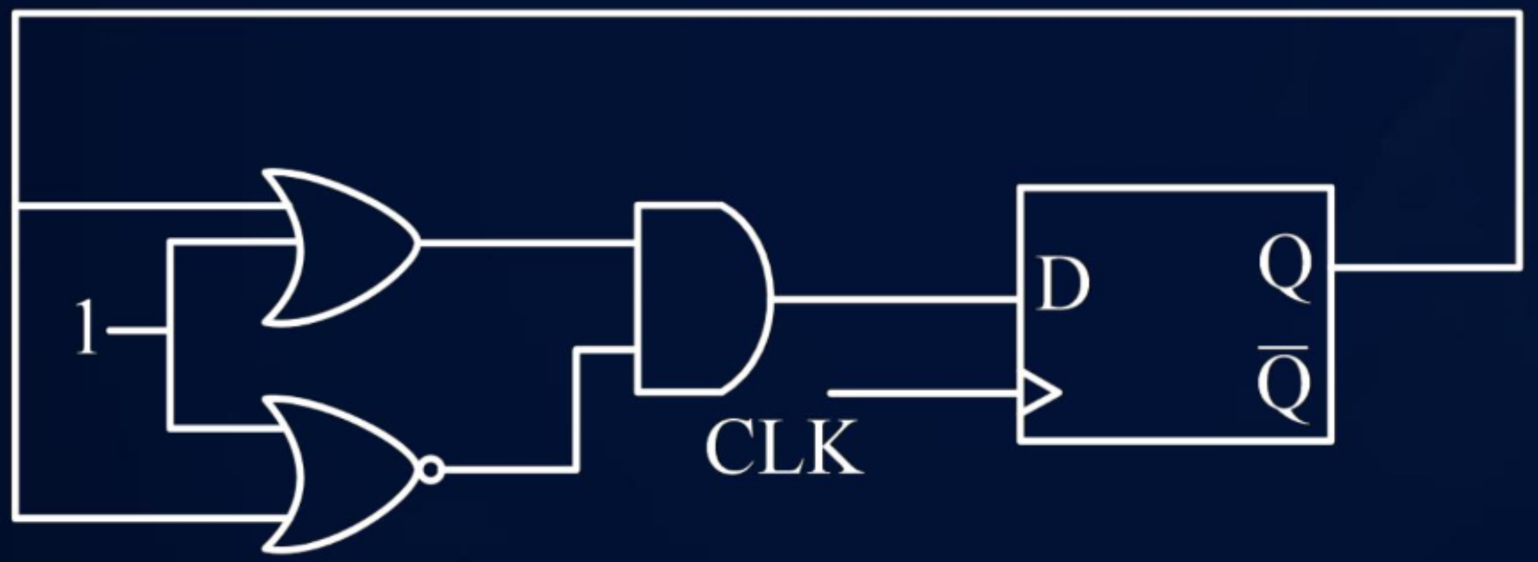
The frequency of the output wave form at 'Q' in kHz is ________

(A) 25

(B) 0

(C) 16.67

(D) 33.33



H.W

Question

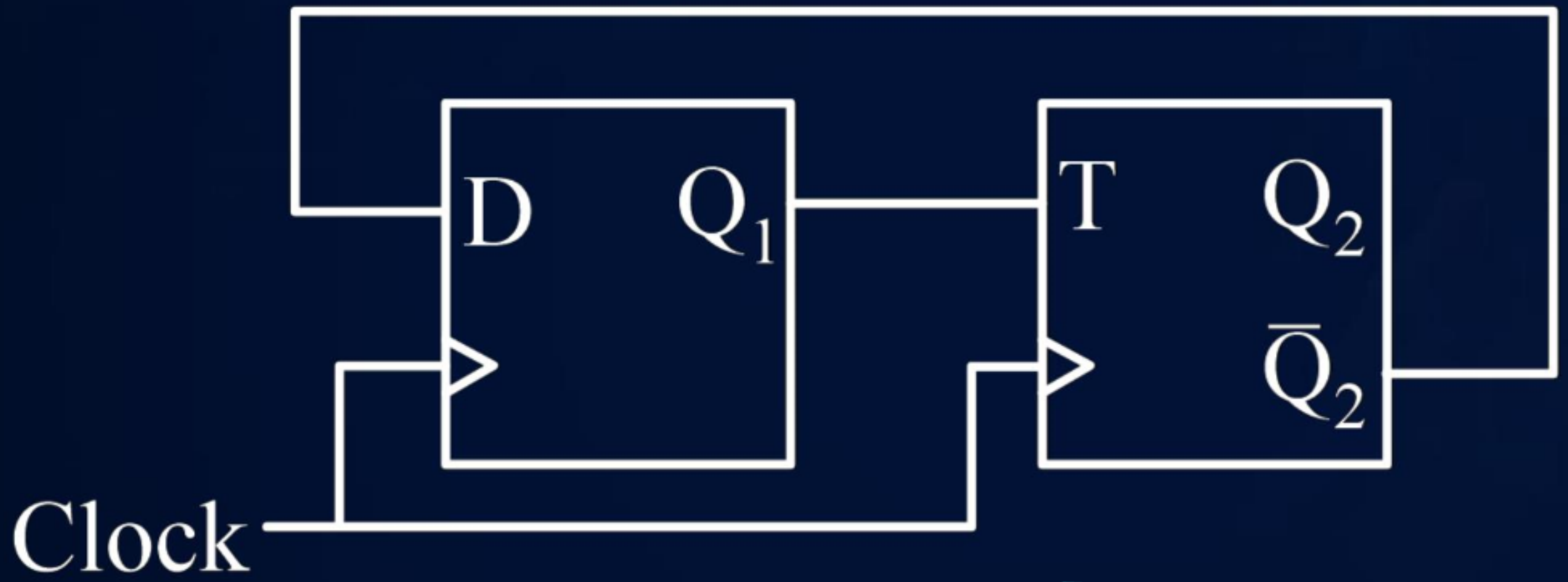
#Q. Assuming $\boxed{Q_1Q_2 = 00}$ initially for the following sequential circuit, then the duty cycle for wave form obtained at $\boxed{\bar{Q}_2 \text{ in \% is}}$

(A) 66.67

(B) 33.33

(C) 50

(D) 25



$Q_1(n+1) = D_1 = \bar{Q}_2(n)$

$Q_2(n+1) = T_2 \oplus Q_2(n) = Q_1(n) \oplus Q_2(n)$

Q_1	Q_2
0	0
1	0
1	1
0	0

$Q_2 \rightarrow T_{off} = 2T_{clk}$

$T_{ON} = 1T_{clk}$

$\bar{Q}_2 \rightarrow T_{ON} = 2T_{clk}$

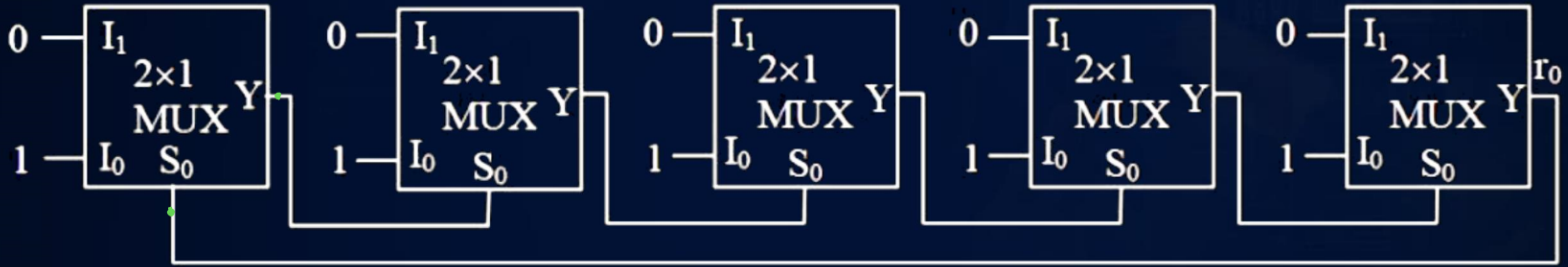
$T_{off} = 1T_{clk}$

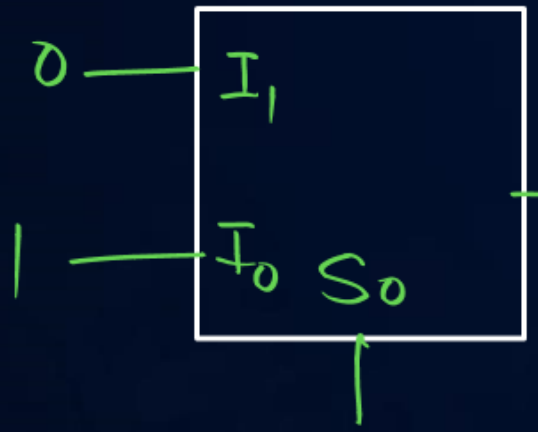
$\%\text{ duty cycle} = \frac{2T_{clk}}{3T_{clk}} \times 100$

$= 66.67\%$

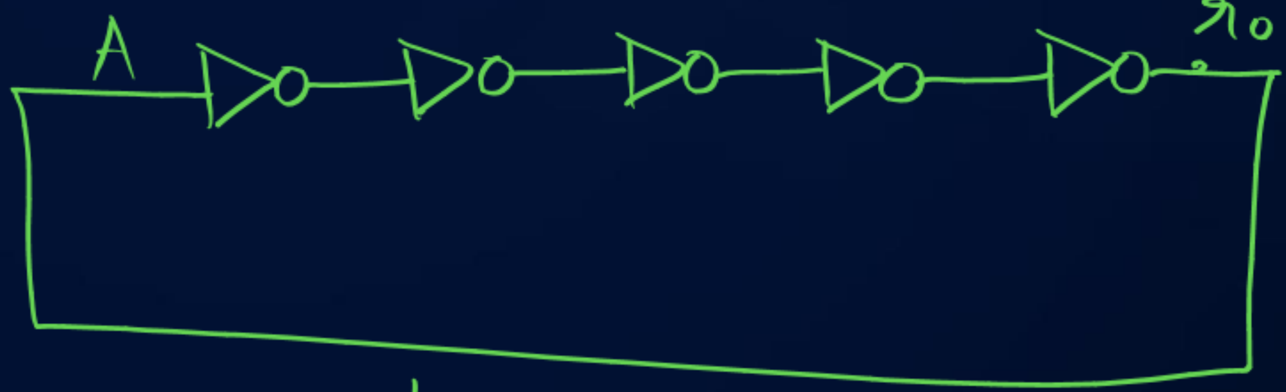
Question

#Q. In the multiplexer-based circuit shown below, the average propagation delay of each MUX is 25 nsec. $\boxed{\text{The frequency of output signal } r_0 \text{ is } \underline{\ \ 4\ \ } \text{ MHz.}}$





$\overline{S_0}\cdot 1 + S_0\cdot 0$
$= \overline{S_0}$



$$f = \frac{1}{2\times 5\times 25\,ns} = \frac{1}{250\,ns} = \frac{10^3}{250}\,MHz = 4\,MHz$$

Question

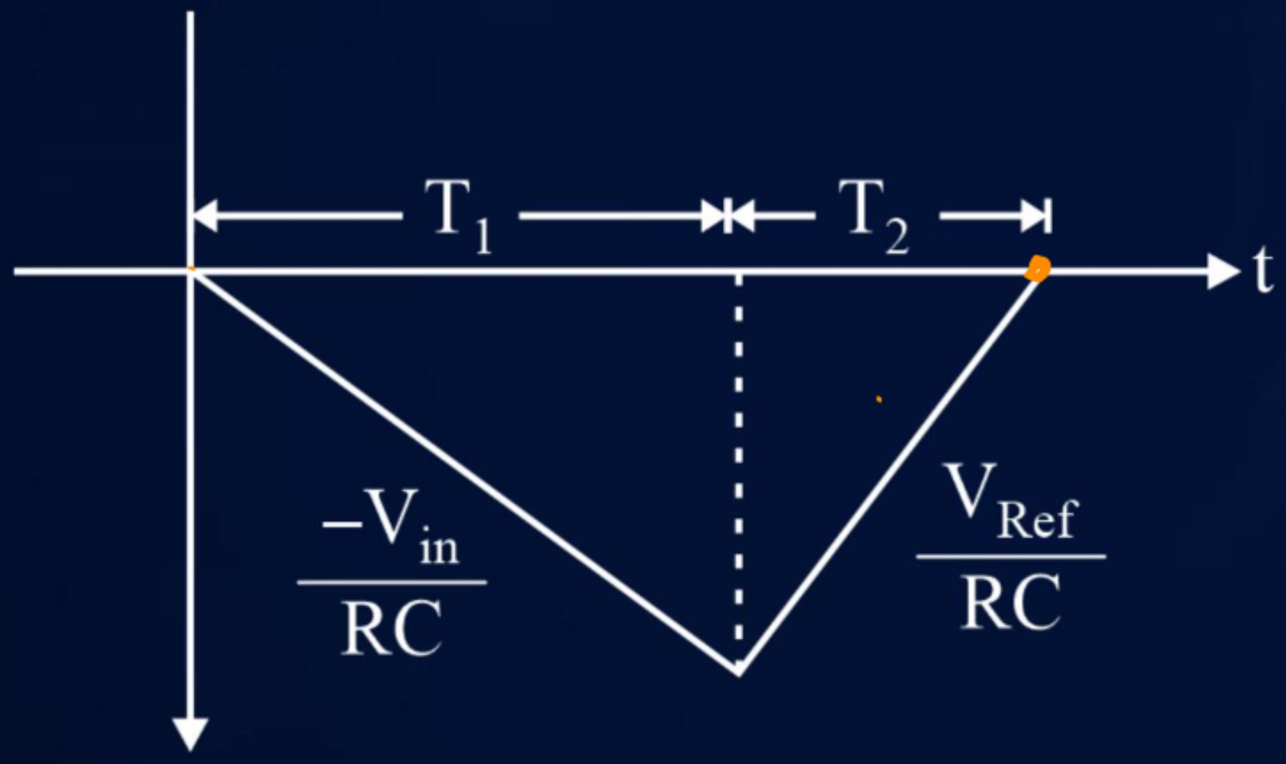
#Q. Consider integrator of dual slope integrating ADC have output as shown below: Assume the value of input voltage (V_i) as 50 mV and reference voltage (V_R) as 100 mV. If the value of T_1 is 50 ms, then value of T_2 will be

A. 50 ms

B. 25 ms

C. 70 ms

D. 12.5 ms



$$\frac{T_2}{T_1} = \frac{V_a}{V_{ref}}$$

$$T_2 = T_1 \times \frac{V_a}{V_{ref}}$$

$$= 50 \times \frac{50}{100}$$

$$= 25\,ms$$

Thank You